

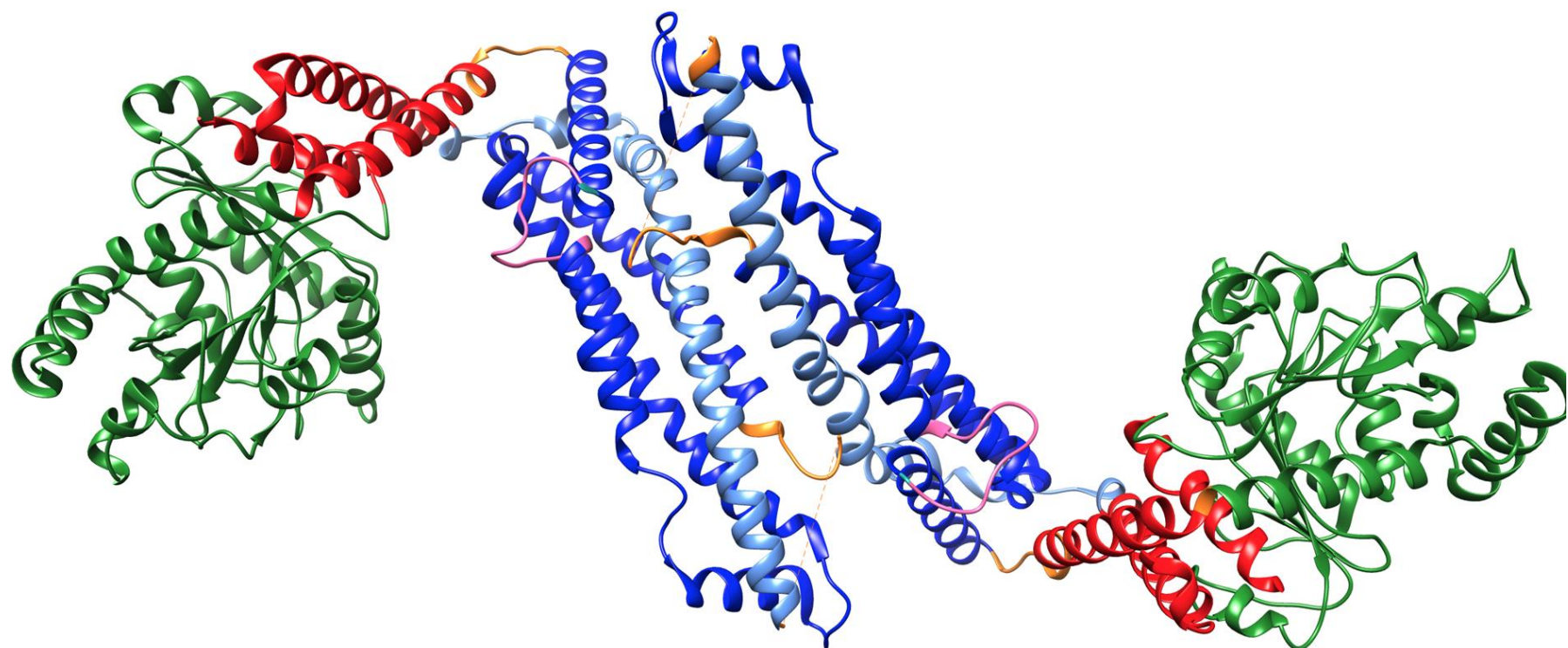
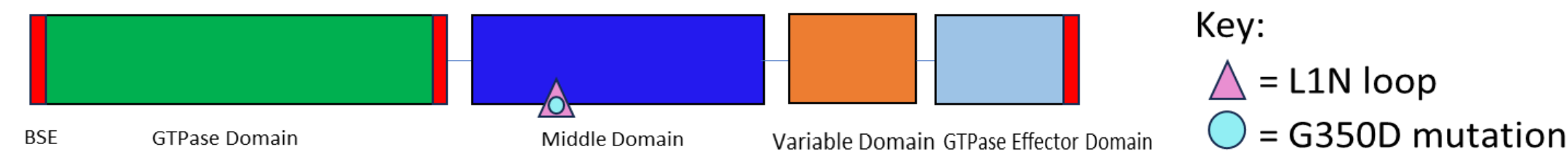
# The functional and structural impacts of the G350D mutation on dynamin-1 related protein

## Abstract

Dynamin-related protein 1 (Drp1) is a key regulator of mitochondrial fission. It assembles into helices that encircle and constrict the mitochondria upon GTP hydrolysis, eventually leading to fission. Mutations in Drp1 can disrupt fission and cause neurodevelopmental and neurodegenerative diseases, though the underlying mechanisms are not fully understood. This study introduces the G350D mutation into Drp1 using bacterial expression constructs to explore its structural and functional impact on the protein. The G350D mutation is located in the Middle Domain region of the protein at one end of the L1N loop - a known hotspot for mutations disrupting assembly. However, the specific influence of this loop on assembly is unknown. Our results revealed that the G350D forms more hexamer structures than the WT in solution, indicating stabilization of larger fission inhibitory complexes. Furthermore, the mutation exhibits increased rates of enzymatic activity ( $3.19 \text{ min}^{-1}$ ) compared to the WT ( $0.13 \text{ min}^{-1}$ ). Negative stain electron microscopy demonstrated that the WT was able to assemble into spirals in the presence of GMP-PCP, whereas the G350D mutation was unable to do so, indicating its assembly defectiveness. Analyzing the effects of specific mutations like G350D is crucial for comprehending Drp1's function and the roles of specific sites like the L1N loop, providing insights into how neurodevelopmental and neurodegenerative diseases occur and offering potential treatment options. Continuing to research mitochondrial dynamics is vital to fully understand the role that mitochondria play in the body and improve human health.

## Background

- Mitochondrial fission is the process by which a singular mitochondrion can split and develop into two or more mitochondria.
- Fission is regulated by dynamin-related protein 1 (Drp1), a large GTPase that forms helices and constricts the mitochondrial membrane upon GTP hydrolysis, eventually leading to fission. (1-2)
- Drp1 self-assembly is crucial for its function. The Middle Domain (MD) region and the GTPase effector domain (GED) of the protein are responsible for controlling assembly.
- Within the MD region, a critical loop (L1N) has been identified as a hotspot for mutations that disrupt protein assembly. L1N stabilizes interactions and promotes assembly, but its specific role and how mutations affect it are not fully understood.
- Mutations in Drp1 can lead to neurodevelopmental and neurodegenerative diseases, though the exact mechanisms by which this occurs remain unclear. (3)
- In this study, the G350D mutation, located in the MD region at one end of the L1N loop, is introduced into Drp1 to analyze its effects on the protein's biochemical structure and function.



**Figure 1.** Diagram depicting the domains of Drp1 chains A and B. The green region represents the GTPase domain, the red region represents the BSE, the dark blue represents the MD, and the light blue represents the GED. The L1N loop is indicated by a pink triangle, with the G350D mutation, located in this loop, marked by a teal circle.

## Goals

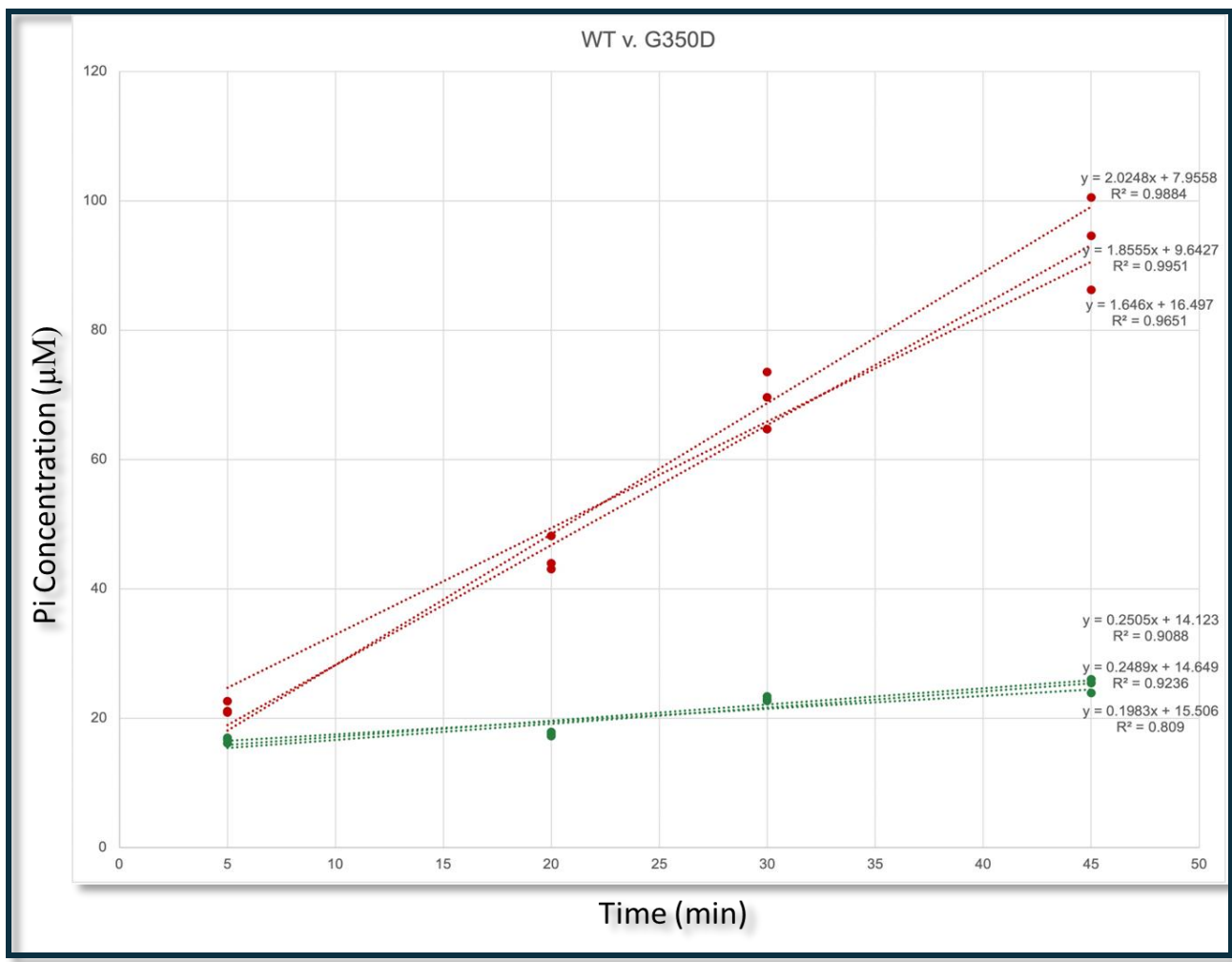
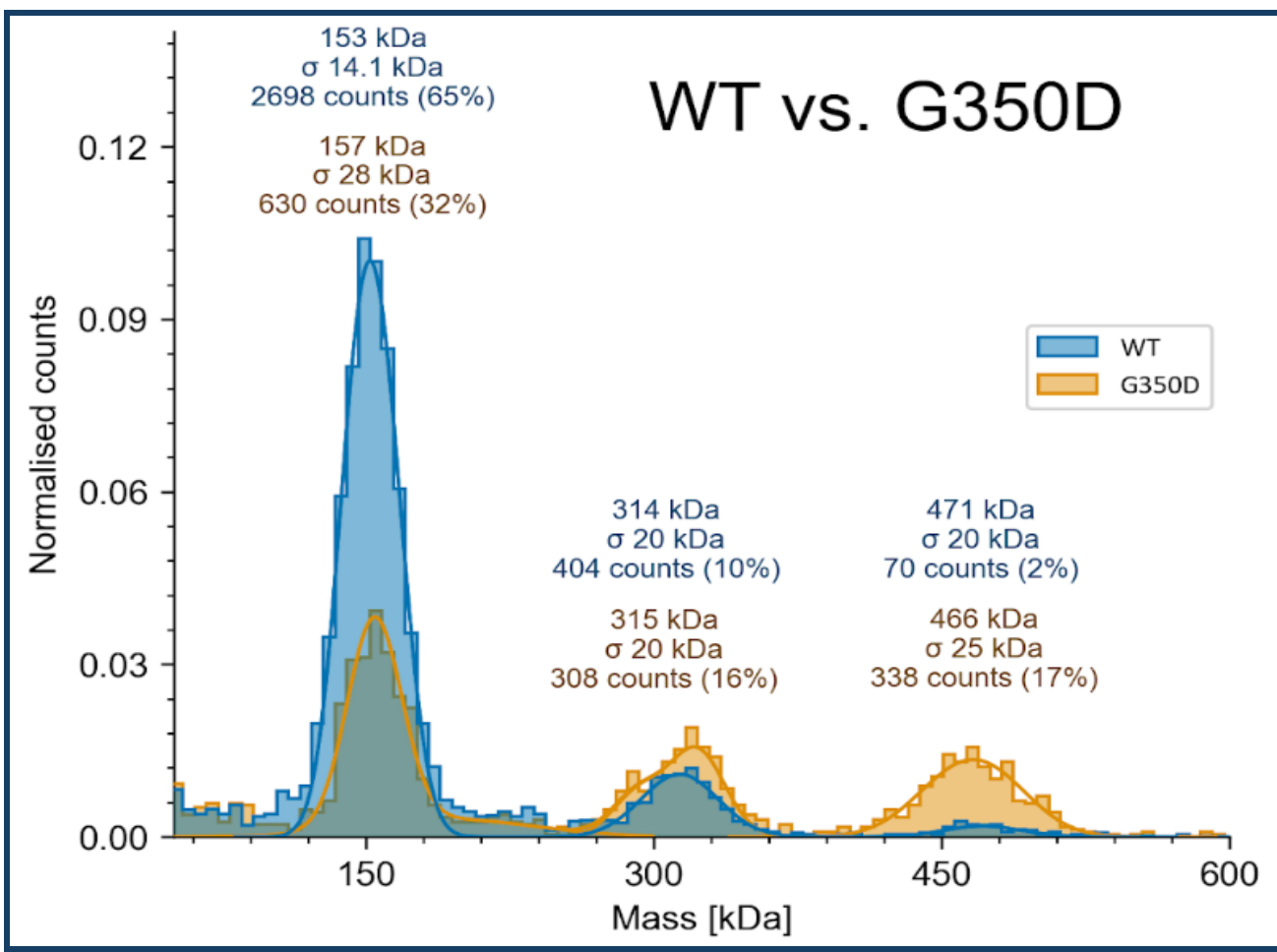
- To investigate the impact of the G350D mutation on Drp1's structure and function.
- To better understand how such mutations impair Drp1 activity and mitochondrial fission.
- To gain insights into how Drp1 mutations can lead to diseases.
- To develop a deeper understanding of the structure and function of the L1N loop and its role in Drp1 assembly.

## Results

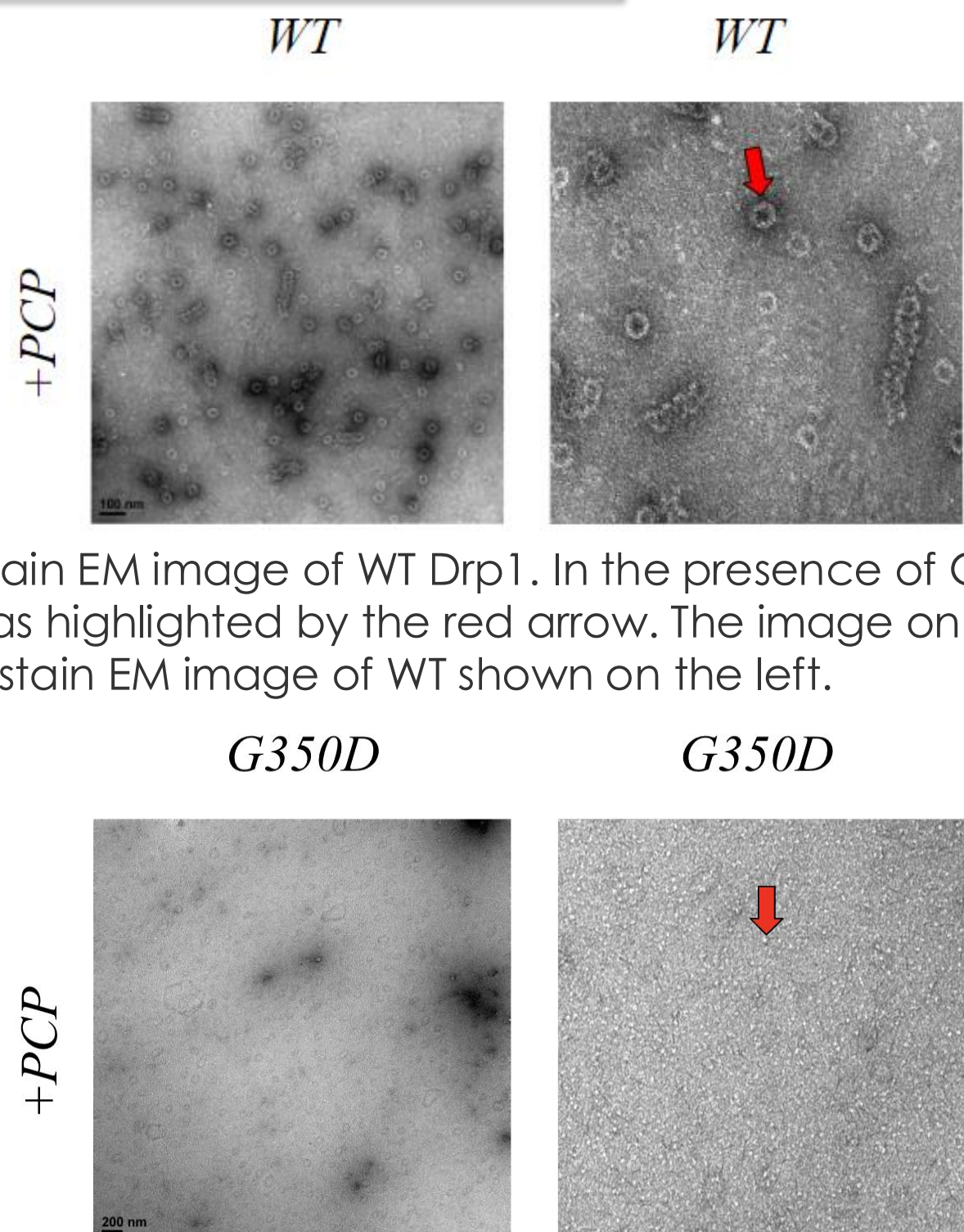
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template sequence Drp1-1 (Drp1-1.xdna)  
TCCNACTTATTACCAAATTTGCCACAGAATATTGTAACACTATTGAAGACACT  
aligned sequence Y322AA-Premixed\_E08.ab1 (Y322AA-Premixed.seq)

**Figure 2.** An image showing the alignment of the Wild Type (WT) Drp1 sequence with the sequence primed by the Y322A primer. This alignment shows that the original Drp1 sequence, which contained GGA, was changed to GAC, highlighted in red.

**Figure 3:** The image depicts the oligomeric forms of both WT and G350D. WT Drp1 predominantly forms dimers, with minimal larger oligomer structures such as hexamers. In contrast, G350D Drp1 shows a more balanced distribution between dimers, tetramers, and hexamers, with significantly more hexamers compared to the WT.



**Figure 4.** The graph displays the GTPase activity of both WT Drp1 (green) and G350D Drp1 (red). The average slope of the G350D Drp1 samples is steeper, showing an average activity of  $4.15 \mu\text{M}$  phosphate/minute/ $\mu\text{M}$  protein or  $4.15 \text{ min}^{-1}$ . In contrast, the average slope of the WT Drp1 samples is lower, with an average activity of  $0.53 \text{ min}^{-1}$ .



**Figure 5.** A negative stain EM image of WT Drp1. In the presence of GMP-PCP, Drp1 assembles into spirals as highlighted by the red arrow. The image on the right is a close-up of the same negative stain EM image of WT shown on the left.

**Figures 6.** A negative stain EM image of G350D Drp1. In the presence of GMP-PCP, the mutant Drp1 does not assemble into spirals. Instead, it remains unassembled and appears as white dots, as highlighted by the red arrow. The image on the right is a close-up of the same negative stain EM image of WT shown on the left.

## Conclusions

- The G350D mutation impairs Drp1's ability to assemble into spiral structures likely due to changes in protein flexibility and charge interactions.
- Mass Photometry revealed that the mutation results in a higher proportion of larger oligomeric states, such as hexamers, indicating a shift away from dimers and suggesting that the mutation limits assembly by restricting the formation of the constriction machinery.
- The mutation increases GTPase activity, resulting in a higher rate of GTP hydrolysis. This is likely due to changes in the L1N loop, which can affect interactions with the GTPase domain and place it in a state optimal state for enzymatic activity.
- This study highlights the impacts of the G350D mutation on Drp1 and emphasizes the importance of the L1N loop in maintaining its own structure and in influencing adjacent regions.
- Continuing this research to further our understanding of the molecular mechanisms of Drp1 mutations provides essential knowledge for developing targeted therapies to mitigate mutation effects, treat neurodevelopmental and neurodegenerative diseases, and restore normal mitochondrial function.

## Materials and Methods

### 1. Sequencing

- Sent purified DNA with G350D mutation for Sanger Sequencing at Azenta Life Sciences.

### 2. Protein Expression

- Cultured BL21 cells with IPTG induction; harvested and stored cells for protein purification.

### 3. Protein Purification

- To purify our mutated protein sequence, we followed the established protocol.

(4)

### 4. Functional Studies

#### ➤ Negative Stain Electron Microscopy (EM)

- Carbon coated mesh grids were placed on the Drp1 protein sample, blotted, then washed with 2% uranyl acetate and blotted again.
- An electron microscope was used to image the grid.

#### ➤ GTPase Assay

- GTP hydrolysis reactions were stopped at different time points to measure the amount of phosphate available.
- After staining the phosphate by malachite green reagent, the plate was imaged by a spectrophotometer to measure the absorbance rate.

#### ➤ Mass Photometry

- The standard protocol was used for measuring light scattering to determine the oligomeric state. (5)

## Future Directions

- Bioenergetics and cell health should be assessed to complement the studies with isolated proteins to see how interactions are affected.
- The mutation should be incorporated into a cell line to observe how impaired fission would impact Drp1's function.
- Future research should also study the impacts of other mutations on Drp1 and incorporate additional assays, such as size exclusion chromatography with multi-angle light scattering.

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